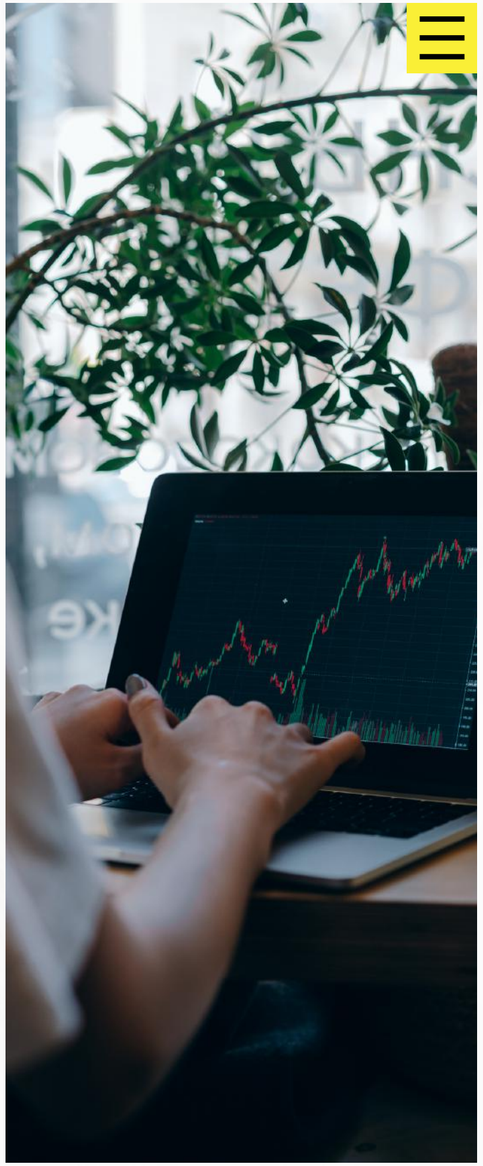
Electric Vehicle In India

Presentation

Presented by Haritha S



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INTRODUCTION

- Electric Vehicles (EVs) are vehicles powered by electricity instead of petrol or diesel.
- EV helps reduce carbon emissions and air pollution, supporting sustainable transportation.
- Government and automobile companies are promoting ev adoption via policies and infrastructure development.
- However, EV adoption varies across regions due to infrastructure availability, affordability, and consumer awareness.
- Understanding EV adoption trends via data analysis helps policymakers and companies make better decisions.

PRESENTED BY:

AARON LOEB

PROJECT OBJECTIVE

- EV vehicles are growing rapidly as a sustainable transportation solution.
- Government and companies need data-driven insights to understand EV adoption pattern.
- However, EV adoption pattern is not uniform across regions, making planning difficult.
- Anayzing EV data helps identitfy infrastructure needs, adoption gaps, and pricing trends.

Tools & Technologies Used

Programming and scripting - Python, SQL Server

Libraries - Pandas, Seaborn,Matplotlib, Streamlit, PyODBC, Logging, Time, OS, Sys, Subprocess

Database - Microsoft SQL Server

Data engineering and Analytics concept - Data cleaning And Validation,
Feature engineering, ETL, Analytics query and dashboarding

TOOLS AND TECHNOLOGY

Business problem

As the EV market grows, **companies and policymakers** struggle to make informed decisions due to limited insights into the EV market.

→ They lack clarity on where EV adoption is highest,
→ which regions have lower adoption,
→ how EV pricing affects affordability,
→ what driving range customers prefer, and
→ how demand varies across different EV range categories.

Because of this, it becomes difficult to plan EV charging infrastructure, design affordable vehicles, and develop targeted strategies to increase EV adoption.

TARGET AUDIENCE

Identify the key audiences for which the project is intended.

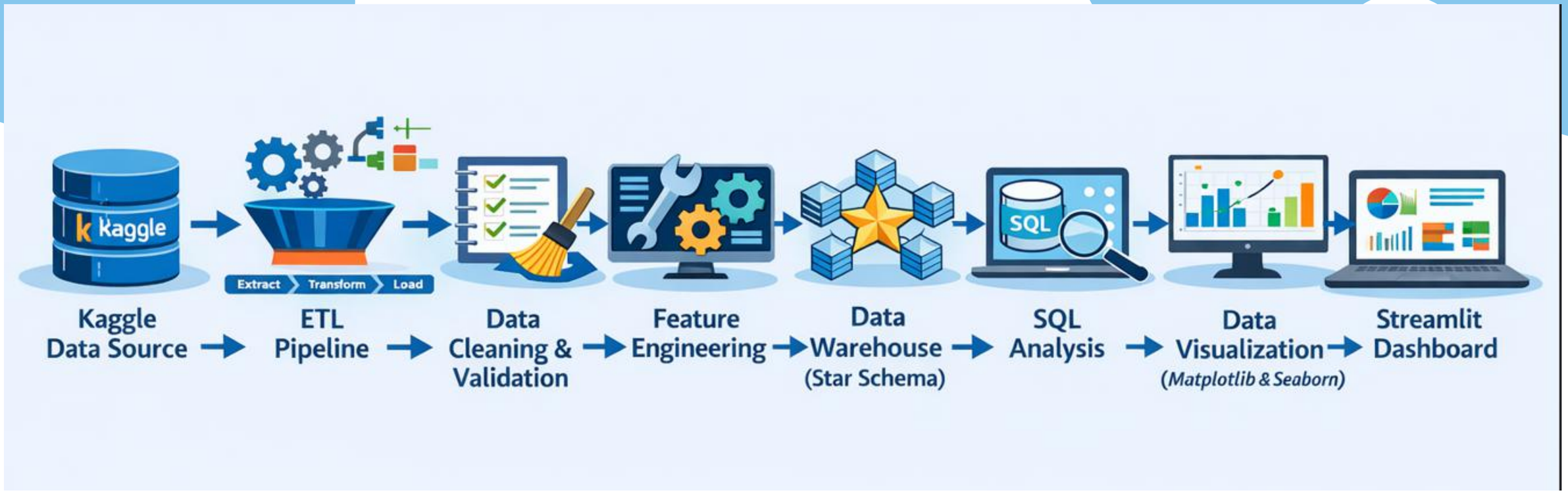
GOVERNEMENT/POLICY MAKERS : To plan EV infrastructure, and policies for improving EV adoption.

EV MANUFACTURING COMPANIES: To understand, market demand, regional adoption patterns and pricing strategies.

INFRASTRUCTURE PLANNERS: To identify locations where charging stations and EV service networks are needed.

MARKET ANALYSIS/RESEARCHERS: To study EV trends and consumer adoption behavior.

SYSTEM ARCHITECTURE



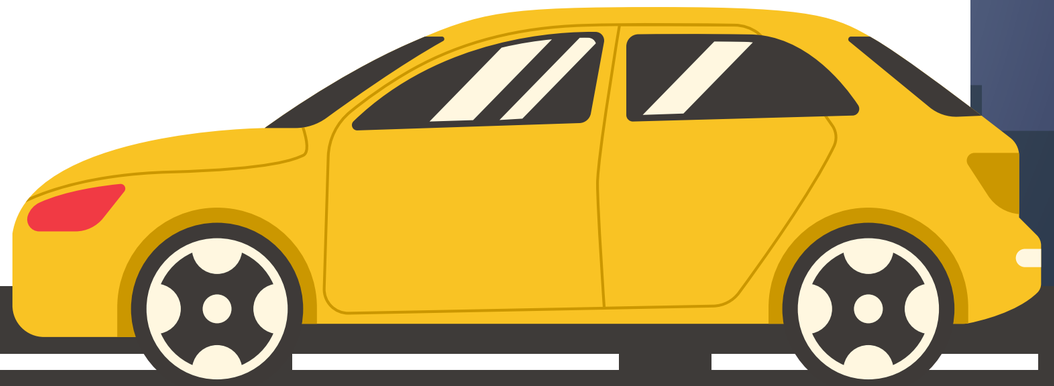
Pipeline Organisation

→ ***D:\Desktop\Electric_Vehicle – Root directory of the EV analytics project***

- ***EV_Datasets*** – Raw CSV datasets used as input data
- ***cleaned_data*** – Cleaned datasets after preprocessing
- ***feature_engineer*** – Datasets with engineered features
- ***Ev_python_file*** – Python scripts for data cleaning, feature engineering, SQL loading, and dashboard
- ***SQL_DB*** – SQL scripts for data warehouse creation and analytical queries
- ***virtualenv*** – Python virtual environment with installed libraries
- ***.idea*** – IDE configuration files

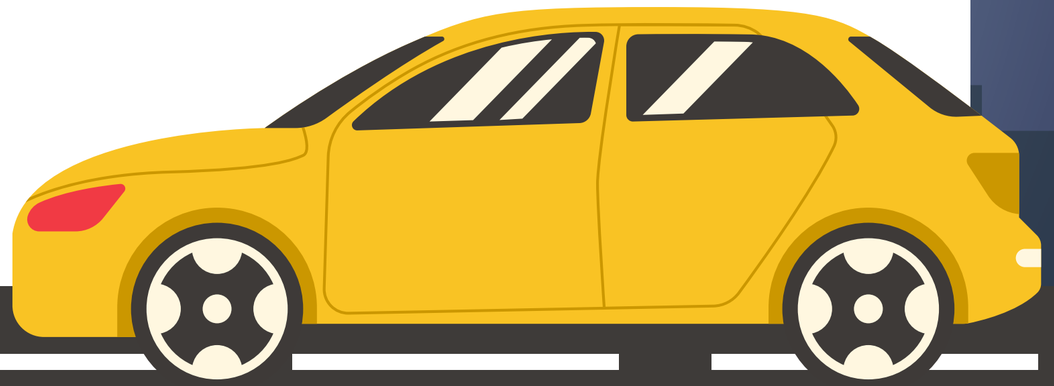
ETL PIPELINE PROCESS

- Extract raw EV datasets
- Transform data through cleaning and preprocessing
- Load processed data into SQL Data Warehouse
- Enable analytics and dashboard visualization



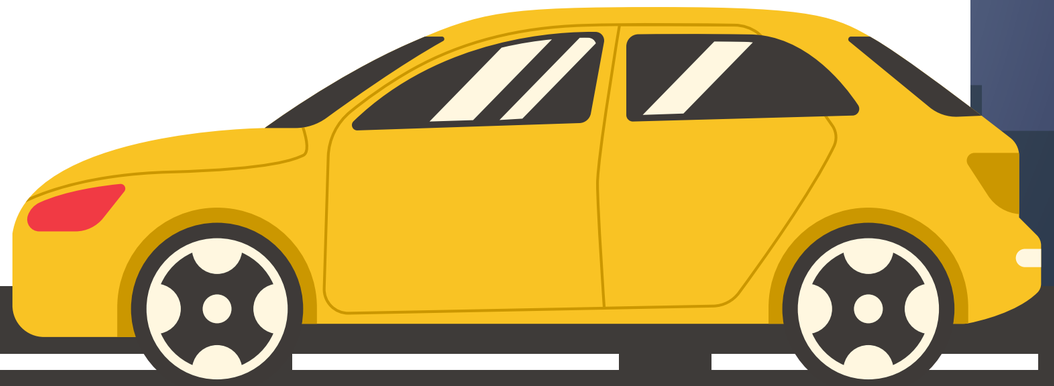
DATA VALIDATION & LOGGING

- Data quality checks implemented
- Missing value handling
- Duplicate record removal
- Logging pipeline execution steps



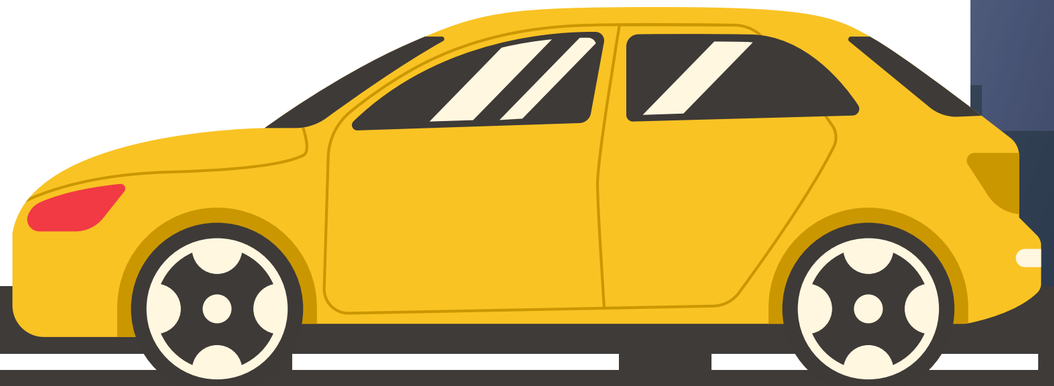
FEATURE ENGINEERING

- Created derived analytical features
- Range category classification
- Premium EV identification
- Vehicle age calculation



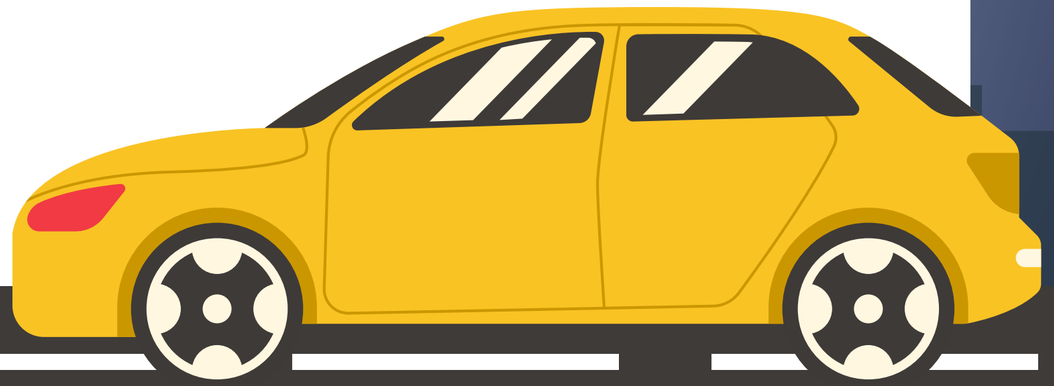
DATA WAREHOUSE DESIGN

- Implemented **Star Schema architecture**
- Fact table for EV metrics
- Dimension tables for descriptive attributes
- Optimized for analytical querie



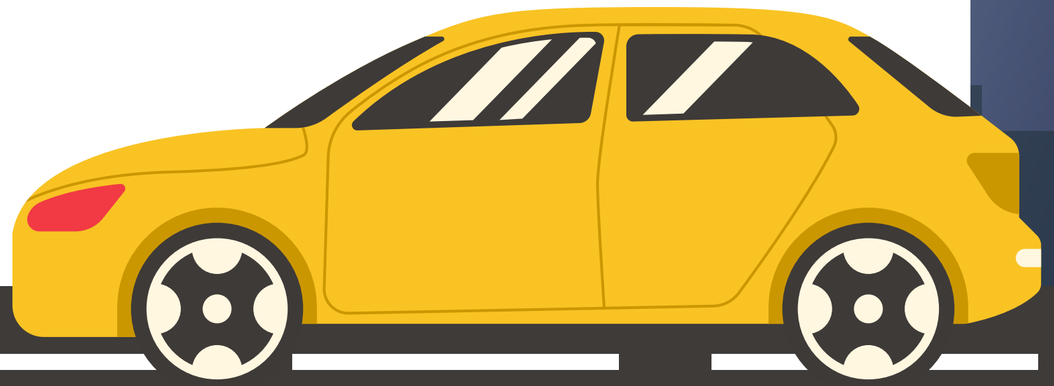
SQL QUERIES FOR ANALYSIS

- Aggregation queries for EV adoption trends
- Region-wise EV distribution analysis
- Metro vs Non-Metro comparisons
- EV price and range insights



DATA ANALYSIS AND VISUALIZATION

- Visual analysis using Python libraries
- Charts created using Matplotlib & Seaborn
- Interactive filtering capabilities
- Dashboard-based exploration



Key insights and result

- EV adoption is **higher in non-metro areas (82.1%)** compared to metro areas (17.9%).
- **South region leads** EV adoption, while the East region shows the lowest adoption.
- The average EV price is around **₹2.7 lakh**, indicating a mid-price market segment.
- **Non-metro areas** (**~298.9 km**) prefer slightly higher driving range EVs when compared to metro areas (~ 295.4 km)
- **Medium-range EVs** dominate the market, balancing affordability and range.

These insights highlight ***regional adoption differences, infrastructure needs, and market trends that influence EV adoption.***

Business Impact

- 82.1% of EV adoption is in non-metro areas
  → ***Indicates the need to expand charging stations and EV infrastructure in non-metro regions.***
- East region has the lowest EV adoption (26,033 vehicles)
  → ***Suggests that policymakers should introduce targeted incentives and awareness programs.***
- Average EV price is around ₹2.7 lakh
- → ***Highlights the importance of affordable EV models to increase adoption.***
- Non-metro range preference (~298.9 km) is slightly higher than metro (~295.4 km)
  → ***Helps manufacturers design higher range EVs for non-metro users***.
- Medium-range and High-range EVs each account for 33.8%
  → ***Indicates balanced customer demand based on price and driving range.***

Future Enhancement

Predictive Modeling:

Use machine learning to forecast EV adoption trends by region or state

Computer Vision:

Analyze images of EV models for features, design, or safety compliance

Infrastructure Planning

Combine EV data with charging station locations to optimize network placement

CONCLUSION

This project developed an **end-to-end EV data analytics pipeline**, including data cleaning, feature engineering, data warehouse design, and interactive visualization.
The **analysis revealed key insights** about EV adoption patterns, regional differences, pricing trends, and driving range preferences.
These insights help policymakers, businesses, and EV manufacturers make data-driven decisions to improve infrastructure planning, market strategies, and support the growth of electric mobility.

Thank you